DATABASE MANAGEMENT SYSTEMS LAB

Lab Report Submitted to

K. R. Mangalam University

for

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

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LAB ASSIGNMENT-1

Design ER Diagrams and Convert Entities & Relationships into Relational Tables

Objective

To design ER diagrams and convert entities and relationships into relational tables using proper schema notation and implement them using SQL.

PART A – COLLEGE DATABASE

Given Relational Schema

STUDENT (USN, SName, Address, Phone, Gender)
SEMSEC (SSID, Sem, Sec)
CLASS (USN, SSID)
SUBJECT (Subcode, Title, Sem, Credits)
IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

ER Diagram Description

Entities:

- STUDENT
- SEMSEC
- SUBJECT
- IAMARKS

Attributes:

STUDENT:

- USN (Primary Key)
- SName
- Address
- Phone (Multivalued attribute)
- Gender

SEMSEC:

- SSID (Primary Key)
- Sem
- Sec

SUBJECT:

- Subcode (Primary Key)
- Title
- Sem
- Credits

IAMARKS:

- USN
- Subcode
- SSID
- Test1
- Test2
- Test3
- FinalIA

Relationships:

1. STUDENT — CLASS — SEMSEC
   (M:N relationship resolved using CLASS)

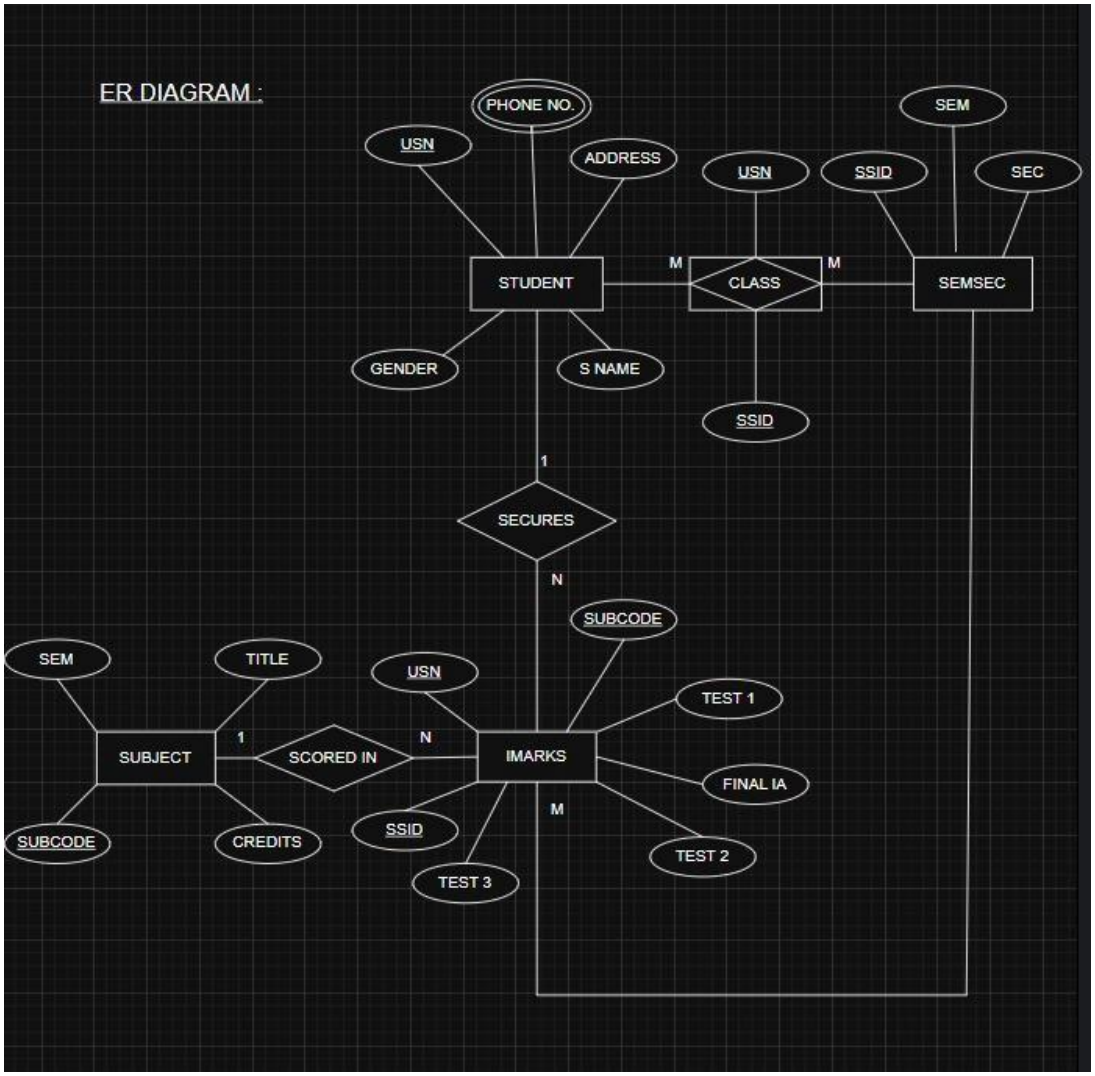
2. STUDENT — IAMARKS (1:M)

3. SUBJECT — IAMARKS (1:M)

4. SEMSEC — IAMARKS (1:M)

Primary keys are underlined in ER diagram.

ER DIAGRAM – PART A



Relational Tables (SQL)

```
-- lab assignment 1

CREATE DATABASE School_DB_POOJA;
USE School_DB_POOJA;

CREATE TABLE STUDENT (
   USN VARCHAR(20) PRIMARY KEY,
   SName VARCHAR(50),
   Address VARCHAR(100),
   Phone VARCHAR(15),
   Gender CHAR(1)
);

CREATE TABLE SEMSEC (
   SSID INT PRIMARY KEY,
   Sem INT,
   Sec CHAR(1)
);
```

```
CREATE TABLE CLASS (
    USN VARCHAR(20),
    SSID INT,
    PRIMARY KEY (USN, SSID),
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);

CREATE TABLE SUBJECT (
    Subcode VARCHAR(10) PRIMARY KEY,
    Title VARCHAR(50),
    Sem INT,
    Credits INT
);

CREATE TABLE IAMARKS (
    USN VARCHAR(20),
    Subcode VARCHAR(10),
    SSID INT,
    Test1 INT,
    Test2 INT,
    Test3 INT,
    FinalIA INT,
    PRIMARY KEY (USN, Subcode, SSID),
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    FOREIGN KEY (Subcode) REFERENCES SUBJECT(Subcode),
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);
```

PART B – COMPANY DATABASE

Given Schema

EMPLOYEE (SSN, Name, Address, Sex, Salary, SuperSSN, DNo)
DEPARTMENT (DNo, DName, MgrSSN, MgrStartDate)
DLOCATION (DNo, DLoc)
PROJECT (PNo, PName, PLocation, DNo)
WORKS_ON (SSN, PNo, Hours)

ER Diagram Explanation

Entities:

- EMPLOYEE
- DEPARTMENT
- PROJECT
- DLOCATION

Relationships:

1. WORKS_ON
    - Between EMPLOYEE and PROJECT
    - M : N
    - Attribute: Hours

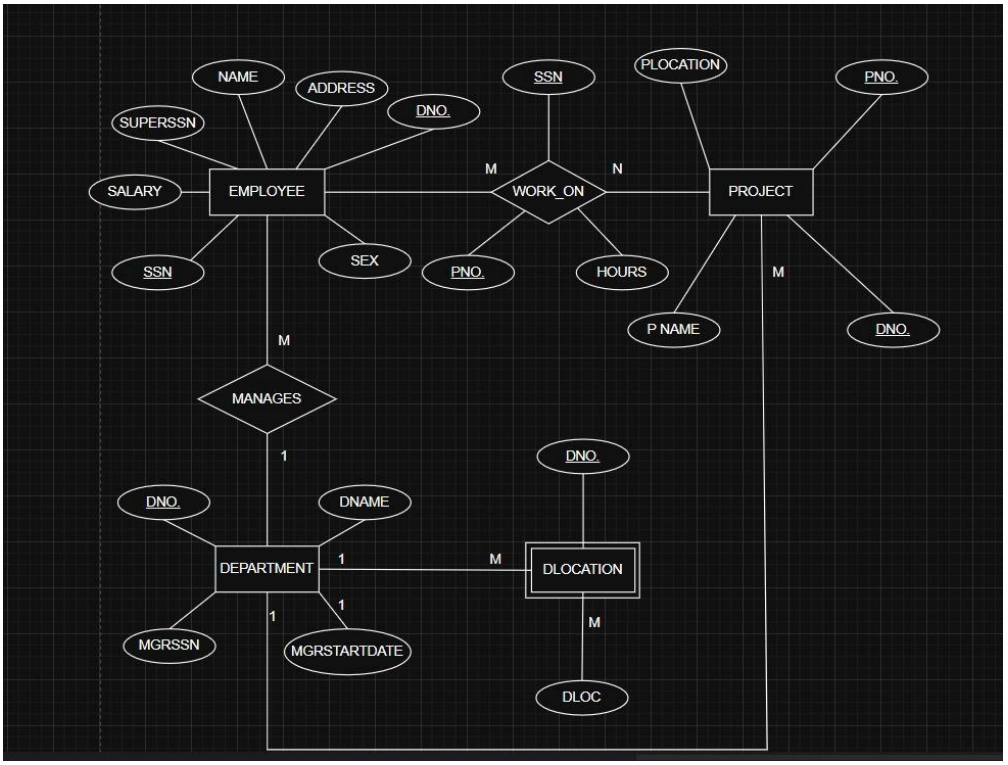
2. MANAGES
    - Between EMPLOYEE and DEPARTMENT
    - 1 : 1
    - Total participation of DEPARTMENT

3. DEPARTMENT — PROJECT
    - 1 : M

4. DEPARTMENT — DLOCATION
    - 1 : M

Primary keys are underlined.

ER DIAGRAM (PART B)



Relational Tables (SQL)

```
CREATE DATABASE company_DB_pooja;
USE company_DB_pooja;

CREATE TABLE DEPARTMENT (
    Dno INT PRIMARY KEY,
    Dname VARCHAR(50),
    Mgrssn VARCHAR(20),
    Mgrstartdate DATE
);

CREATE TABLE EMPLOYEE (
    SSN VARCHAR(20) PRIMARY KEY,
    Name VARCHAR(50),
    Address VARCHAR(100),
    Sex CHAR(1),
    Salary INT,
    Superssn VARCHAR(20),
    Dno INT,
    FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
);
```

PART C – CUSTOMER SALE SCHEMA

Customer (cust_id, cust_name)
Item (item_id, item_name, price)
Sale (bill_no, bill_date, cust_id, item_id, qty_sold)

ER Explanation

Customer → Sale = 1 : M
Item → Sale = 1 : M

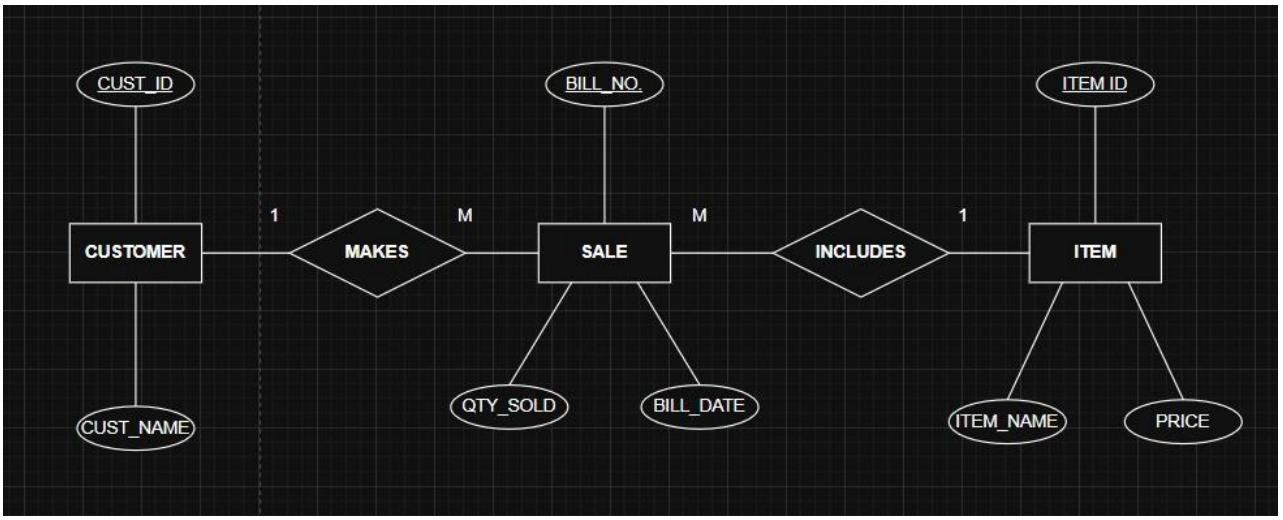
Foreign Keys:

- cust_id references Customer
- item_id references Item

Referential Integrity:

- Sale must have valid customer
- Sale must have valid item

ER DIAGRAM (PART C)



Relational Tables (SQL)

```
CREATE DATABASE Sales_DB_pooja;
USE Sales_DB_pooja;

CREATE TABLE CUSTOMER (
    cust_id INT PRIMARY KEY,
    cust_name VARCHAR(50)
);
```

```
CREATE TABLE ITEM (
   item_id INT PRIMARY KEY,
   item_name VARCHAR(50),
   price INT
);

CREATE TABLE SALE (
   bill_no INT PRIMARY KEY,
   bill_date DATE,
   cust_id   INT,
   item_id   INT,
   qty_sold INT,
   FOREIGN KEY (cust_id) REFERENCES CUSTOMER(cust_id),
   FOREIGN KEY (item_id) REFERENCES ITEM(item_id)
);
```

DDL QUESTIONS

1. Add Email column

```
ALTER TABLE STUDENT ADD Email VARCHAR(50);
```

2. Apply NOT NULL constraints

```
ALTER TABLE EMPLOYEE MODIFY Salary INT NOT NULL;
ALTER TABLE ITEM MODIFY price INT NOT NULL;
```

3. Rename S Name to Student Name

```
ALTER TABLE STUDENT RENAME COLUMN S Name TO Student Name;
```

4. Drop DLOCATION

```
DROP TABLE DLOCATION;
```

DML QUESTIONS

1. Insert 10 tuples (example)

```
INSERT INTO CUSTOMER VALUES (1,'Ram');
INSERT INTO CUSTOMER VALUES (2,'Amit');
INSERT INTO CUSTOMER VALUES (3,'Kir');
INSERT INTO CUSTOMER VALUES (4,'Sam');
INSERT INTO CUSTOMER VALUES (5,'Aamy');
INSERT INTO CUSTOMER VALUES (6,'Manav');
INSERT INTO CUSTOMER VALUES (7,'Ravi');
INSERT INTO CUSTOMER VALUES (8,'Divya');
INSERT INTO CUSTOMER VALUES (9,'Niaha');
INSERT INTO CUSTOMER VALUES (10,'Neha');
```

2. List bills of current date

```
SELECT s.bill_no, c.cust_name, i.item_name
FROM SALE s
JOIN CUSTOMER c ON s.cust_id = c.cust_id
JOIN ITEM i ON s.item_id = i.item_id
WHERE s.bill_date = CURDATE();
```

---

3. Total bill amount

```
SELECT bill_no, qty_sold * price AS Total_Amount
FROM SALE s
JOIN ITEM i ON s.item_id = i.item_id;
```

---

4. Customers who purchased items above 200

```
SELECT DISTINCT c.cust_name
FROM SALE s
JOIN CUSTOMER c ON s.cust_id = c.cust_id
JOIN ITEM i ON s.item_id = i.item_id
WHERE price > 200;
```

---

5. Count products per customer

```
SELECT cust_id, COUNT(item_id)
FROM SALE
GROUP BY cust_id;
```

---

6. Products purchased by Cust_id = 5

```
SELECT item_id
FROM SALE
WHERE cust_id = 5;
```

---

7. Create View (Bill Details)

```
CREATE VIEW BILL_VIEW AS
SELECT s.bill_no, s.bill_date, s.cust_id, s.item_id,
    i.price, s.qty_sold,
    (s.qty_sold * i.price) AS amount
FROM SALE s
JOIN ITEM i ON s.item_id = i.item_id;
```